

Phishing Campaign Profiling

Data Mining Project Proposal

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Problem Statement:

- Phishing attacks continuously adapt social-engineering lures to bypass static rule systems.

Primary Project Objective:

- Build an end-to-end unsupervised pipeline to segment email campaigns into tactical archetypes and isolate evasive anomalies.

Research Questions:

1. What specific textual features, indicators, or structural email attributes distinguish standard promotional spam from dangerous phishing attempts?
2. Can structural characteristics (sender domain, routing paths, link destinations) identify distinct clusters or campaign structures among malicious emails?
3. What anomalous features mark high-risk/unusually evasive phishing emails, and how well do unsupervised outlier techniques flag them compared to known ground-truth labels?

Datasets: Enron and Ling.

- **Unit of Analysis:** A single email message (*Subject* and *Body*).

Source	Safe (0)	Malicious (1)	Total Records	Malicious %
Enron	15,791	13,976	29,767	46.95%
Ling-Spam	2,401	458	2,859	16.02%
Total Combined	18,192	14,434	32,626	44.24%

Data Quality & Scope Controls:

- **Downsampling:** Downsampling records for computationally expensive algorithms.
- **Biased:** There is more data from Enron than Ling meaning some patterns could appear for one dataset and create bias.

1. Data Handling

- HTML stripping, missing value imputation.

2. Text Mining

- e.g. Tokenization, stop-words.

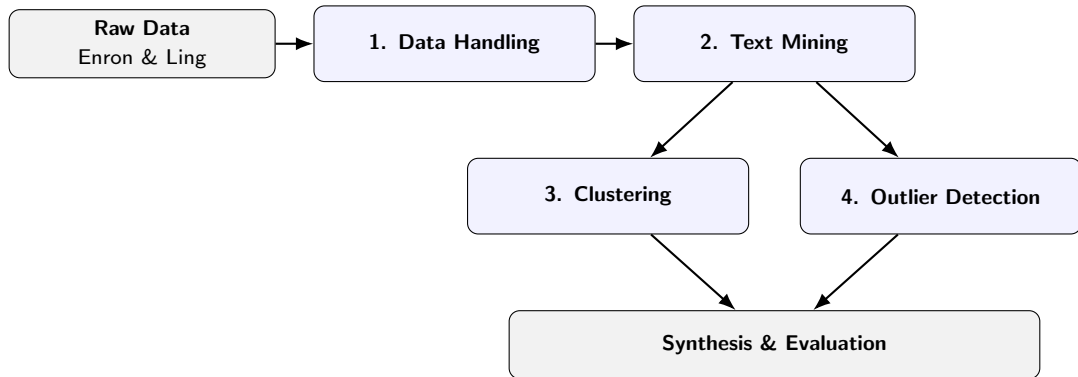
3. Clustering

- Grouping emails into campaign archetypes.

4. Outlier Detection

- Detecting low-volume, stealthy spear-phishing.

Initial Pipeline Diagram



Expected Evaluation Approach

- **Data Handling:** Quality before/after metrics.
- **Text Mining:** Topic coherence scores and vocabulary stability.
- **Clustering:** Evaluation using different metrics.
- **Outlier Detection:** Overlap/stability analysis.

- **Risk 1: High Sparsity**
 - *Mitigation:* Apply vocabulary frequency filtering.
- **Risk 2: Computational Bottlenecks**
 - *Mitigation:* Execute the evaluation pipeline on a subset.
- **Risk 3: Unbalanced Classes and Ground-Truth Ambiguity**
 - *Mitigation:* Unsupervised pattern discovery.